

LAVANYA CHANDEL

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Summary

Second-year B.Tech IT student at IIIT Jabalpur with hands-on experience shipping production web applications and a growing focus on systems programming. Built and deployed a full-stack platform used by a live business, and independently implemented concurrent systems in C++. Active competitive programmer with experience across core DSA problem domains.

Education

Indian Institute of Information Technology, Jabalpur

Aug 2024 – Aug 2028

Bachelor of Technology

Jabalpur, Madhya Pradesh

Coursework: Data Structures & Algorithms, Discrete Mathematics, Probability & Statistics, Linear Algebra

Technical Skills

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, C

Frameworks & Libraries: Next.js, React, Node.js, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib

Tools: Git, GitHub, Vercel, VS Code, Android Studio, MySQL

Areas: Data Structures & Algorithms, Concurrent Systems, Full-Stack Development

Professional Experience

Frontend Developer Intern

May 2025 – Jul 2025

Vedanta Home Interior

Jaipur, Rajasthan

- Architected and deployed VLOOMS (vlooms.in) from scratch using Next.js 15 App Router + TypeScript, configuring Git-based CI/CD on Vercel for zero-downtime production delivery.
- Integrated WhatsApp Business API for real-time client-to-wholesale lead capture, improving communication efficiency for a live fabric business.
- Built mobile-responsive UI with scroll animations via IntersectionObserver API; ensured cross-device and cross-browser consistency through systematic testing.

Projects

Thread Pool Implementation | C++17 | github.com/lavcharms/cpp-thread-pool

- Built a fixed-size thread pool from scratch in C++ managing concurrent task execution across N worker threads using a shared task queue.
- Implemented mutex-based synchronization and condition variables to eliminate busy-waiting; threads sleep when idle and wake only on task arrival.
- Used `std::atomic` for lock-free completion tracking and `std::function` for type-erased task dispatch, supporting arbitrary callables.

VLOOMS Marketing Platform | Next.js 15, TypeScript, React, Vercel | github.com/lavcharms/vlooms

- Architected component structure using Next.js 15 App Router with TypeScript strict mode, defining reusable interfaces and types across the codebase.
- Implemented dynamic fabric catalogue with live category filtering via React state management, handling real product data across multiple categories.
- Owned full deployment lifecycle from local dev to production using Git version control and Vercel CI/CD with custom domain configuration.

Classroom Availability Analyser | Python, Algorithmic Scheduling

Aug 2025

- Designed a data-driven scheduling system using constraint-based algorithmic logic to compute real-time classroom availability across dynamic timetable configurations.
- Placed 1st out of 70+ participants at Can You Hack It! (all-girls hackathon, IIIT Jabalpur); delivered a working demo end-to-end under a timed constraint.

Competitive Programming & Achievements

Competitive Programming

Apr 2025 – Present

Codeforces & CodeChef

IIIT Jabalpur

- Solved **130+** problems on [CodeChef](https://codechef.com) (rating: **885**) and [Codeforces](https://codeforces.com) (rating: **1136**); actively competing in rated contests.
- Practicing medium/hard problems in graphs, dynamic programming, and greedy algorithms; targeting consistent rating growth.